Exact Structure in Accelerated Collatz Dynamics: Valuation Laws, Orbit Identities, and a Machine-Certified Descent Atlas

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July 2026

Abstract

We collect exact, unconditional structure for the accelerated Collatz map $\mathrm{step}(x) = (3x+1)/2^{\mathrm{v}_2(3x+1)}$ on odd integers: a run-length law identifying $\mathrm{v}_2(x+1)-1$ as the exact number of initial valuation-one steps; an up/down dichotomy and a 2-adic resistance-descent lemma for the two-sided map $3x\pm1$; exact orbit identities and a closed form for the logarithmic defect; and a machine-certified atlas of descent classes (197 classes / 1399 theorems, Lean 4, kernel-only axioms). We prove two limitative results: for no modulus M is $\mathrm{v}_2(3x+1)$ a function of x mod M, so no finite automaton on residues captures the dynamics — and the atlas method reaches "almost all n" but provably never all n. **This paper does not prove the Collatz conjecture and makes no such claim**; it delimits what the present methods can and cannot do.

MSC: 11B83, 11A07 (Congruences), 37P05 (Arithmetic dynamics), 68V20 (Formalisation of mathematics).

1 Setting and scope

On the odd integers put $\mathrm{step}(x) = (3x+1)/2^{\mathrm{v}_2(3x+1)}$, where v_2 is the 2-adic valuation. For an orbit reaching 1 with odd iterates $a_1 = n, \ldots, a_A$ and valuations $v_i = \mathrm{v}_2(3a_i+1)$, write $S = \sum_i v_i$ and $S_i = \sum_{j\le i} v_j$ ($S_0 = 0$).

Each result carries a status: **[P]** proved; **[P-Lean]** additionally machine-checked in Lean 4 with kernel-only axioms; **[V]** exactly verified on a stated range with 0 exceptions; **[C]** conjecture.

Companion notes. Two results used here are of independent interest and are treated separately: the equidistribution of the valuations [1] and the characterisation of the bases a for which $ax+1$ has closed-form orbits [2].

2 A run-length law

Theorem 2.1 (K_0 law, **[P]**). *For odd n set $k(n) = \mathrm{v}_2(n+1)-1$. The orbit of n begins with* exactly $k(n)$ *consecutive steps of valuation* 1*; and if $k(n) = 0$ then* $\mathrm{step}(n) < n$.

Proof. Write $n = 2^{k+1}m - 1$ with m odd, $k = k(n)$. Then $3n+1 = 2(3\cdot 2^k m - 1)$ and $3\cdot 2^k m - 1$ is odd exactly when $k \ge 1$; hence $v = 1$ and $\mathrm{step}(n) = 2^k(3m) - 1$, of the same shape with k reduced by one. After k steps the parameter is 0, where $4 \mid 3n+1$ and $\mathrm{step}(n) \le (3n+1)/4 < n$ for $n > 1$. □

Verified with 0 exceptions for all odd $n < 2\times 10^5$. The proof uses only that 3 is odd, so it holds verbatim for $ax+1$ with a odd.

3 The two-sided map $3x \pm 1$

Each odd x now has two children, odd_part$(3x+1)$ and odd_part$(3x-1)$.

Observation 3.1 (Up/down dichotomy, **[V]**)**.** For every odd $n \geq 3$, $3n+1$ and $3n-1$ are consecutive even numbers, so exactly one is $\equiv 0 \pmod 4$ (valuation ≥ 2) and the other $\equiv 2 \pmod 4$ (valuation 1). Hence one child is $\frac{3n\pm1}{2} > n$ (the *up* child, factor $\approx \frac{3}{2}$) and the other is $< \frac{3n+1}{4} < n$ (the *down* child, factor $3/2^k$, $k \geq 2$). Consequently $\mathrm{v}_2(3n+1) + \mathrm{v}_2(3n-1) \geq 3$, never 2.

Verified with 0 exceptions for all odd $n \leq 2 \times 10^5$.

Theorem 3.2 (2-adic resistance descends, **[P]**)**.** *Let $r(n) = \max(\mathrm{v}_2(n-1), \mathrm{v}_2(n+1))$ be the 2-adic distance to ± 1 (so $r \geq 2$ for odd n). For every odd n with $r(n) = r \geq 4$, the two children have resistance* exactly $r-1$ *(up child) and $r-2$ (down child). Hence r strictly decreases while $r \geq 4$: from resistance r the walk returns to $r \leq 3$ within r steps.*

Proof. Take $n = 1 + 2^r m$, m odd (the case $n \equiv -1$ is symmetric). Then $3n+1 = 4(1 + 3 \cdot 2^{r-2} m)$, whose odd part d has $d - 1 = 3 \cdot 2^{r-2} m$, so $\mathrm{v}_2(d-1) = r-2$ while $\mathrm{v}_2(d+1) = 1$; hence $r(d) = r-2$. And $3n - 1 = 2(1 + 3 \cdot 2^{r-1} m)$, whose odd part u has $\mathrm{v}_2(u-1) = r-1$ and $\mathrm{v}_2(u+1) = 1$; hence $r(u) = r-1$. □

Verified with 0 exceptions for all odd $n \leq 2 \times 10^6$.

4 Orbit identities

Proposition 4.1 (Exact orbit identity, **[P]**)**.** $2^S = 3^A n + \sum_{i=0}^{A-1} 3^{A-1-i} 2^{S_i}$.

The classical relation behind the 3^A-versus-2^S bookkeeping (cf. [3]); restated because the next result depends on it. Verified with 0 exceptions for odd $n < 2 \times 10^4$.

Proposition 4.2 (Logarithmic defect in closed form, **[P]**)**.** *With $\delta := S - A \log_2 3 - \log_2 n$, we have $\delta = \sum_{i=1}^{A} \log_2(1 + \frac{1}{3a_i})$; in particular $\delta > 0$ always.*

Proof. Take $\log_2$ of $2^{v_i} a_{i+1} = 3a_i(1 + \frac{1}{3a_i})$ and sum; the $\log_2 a_i$ telescope. □

Verified to $|\cdot| < 1.5 \times 10^{-14}$ for odd $n < 4000$.

Proposition 4.3 (Predecessor law mod 3, **[P]**)**.** *Let $a_{i-1} \mapsto a_i = \mathrm{step}(a_{i-1})$ be consecutive odd terms, $v_{i-1} = \mathrm{v}_2(3a_{i-1}+1)$. Then*

$$\begin{aligned} a_i \equiv 0 \pmod 3 &\implies a_i \text{ has no odd predecessor}, \\ v_{i-1} \text{ even} &\iff a_i \equiv 1 \pmod 3, \\ v_{i-1} \text{ odd} &\iff a_i \equiv 2 \pmod 3. \end{aligned}$$

Proof. Inverting the step, $a_{i-1} = (2^{v_{i-1}} a_i - 1)/3$ must be a positive integer, so $2^{v_{i-1}} a_i \equiv 1 \pmod 3$. As $2 \equiv -1 \pmod 3$ this reads $(-1)^{v_{i-1}} a_i \equiv 1 \pmod 3$. For $a_i \equiv 0$ there is no solution; for $a_i \equiv 1$ it forces $(-1)^{v_{i-1}} = 1$, i.e. v_{i-1} even; for $a_i \equiv 2 \equiv -1$ it forces v_{i-1} odd. □

Verified on 4,041,077 consecutive pairs (all odd $n < 2 \times 10^5$), 0 violations.

Corollary 4.4 (Last-pair parity, **[P]**)**.** *Let a_A be the last odd term,* $\mathrm{step}(a_A) = 1$*. Then v_A is even and $a_A = (2^{v_A} - 1)/3$. Writing $v_A = 2m$, we have $a_A = (4^m - 1)/3 = \sum_{i<m} 4^i \equiv m \pmod 3$, and hence for $A \geq 2$:*

$$v_A \equiv 2 \pmod 6 \Rightarrow v_{A-1} \text{ even}, \qquad v_A \equiv 4 \pmod 6 \Rightarrow v_{A-1} \text{ odd},$$

while $v_A \equiv 0 \pmod 6$ forces $a_A \equiv 0 \pmod 3$, so a_A has no predecessor and $A = 1$.

Proof. $3a_A + 1 = 2^{v_A}$ needs $2^{v_A} \equiv 1 \pmod 3$, i.e. v_A even. The rest is Proposition 4.3 with $a_A \equiv m \pmod 3$, since $m \equiv 1, 2, 0 \pmod 3$ corresponds to $v_A \equiv 2, 4, 0 \pmod 6$. □

This settles what was previously recorded only as an empirical observation, and explains it: the "modulus 6" is modulus 3 in disguise, the factor 2 coming from $v_A = 2m$. The admissible last terms are $a_A \in \{1, 5, 21, 85, 341, \dots\} = (4^m - 1)/3$, and those with $3 \mid a_A$ (i.e. $21, 1365, \dots$) are orbit *starts*: nothing maps to them.

5 No finite-residue reduction

Proposition 5.1 (**[P]**). *For every modulus M, the valuation $\mathrm{v}_2(3x+1)$ is* not *a function of $x \bmod M$.*

Proof. Write $M = 2^s M'$ with M' odd and let $c = -3^{-1} \bmod 2^{s+1}$, which is odd. Every $x \equiv c \pmod{2^s}$ has $\mathrm{v}_2(3x+1) \geq s$, but the exact value is decided by the binary digits of x above position s — which $x \bmod M$ does not see. By the Chinese Remainder Theorem pick odd $x \equiv y \pmod M$ with $3x + 1 \equiv 2^s \pmod{2^{s+1}}$ and $3y + 1 \equiv 0 \pmod{2^{s+1}}$; the systems are compatible since they constrain the 2-part and the odd part M' at coprime places. Then $\mathrm{v}_2(3x+1) = s < s + 1 \leq \mathrm{v}_2(3y+1)$. □

So $\mathrm{v}_2(3x+1) \geq s$ is decided by $x \bmod 2^s$ but $\mathrm{v}_2(3x+1) = s$ exactly is not: the valuation sequence requires unbounded 2-adic precision and the dynamics is *not* a finite automaton on residues. At $M = 24$, among the 2000 odd $x < 4000$: 500 have $\mathrm{v}_2(3x+1) \neq \mathrm{v}_2(3(x+24)+1)$ (least witness $x = 5$, $y = 29$: $\mathrm{v}_2(16) = 4 \neq 3 = \mathrm{v}_2(88)$), and 1959 have $\mathrm{step}(x) \not\equiv \mathrm{step}(x+24) \pmod{24}$ (least witness $x = 1$, $y = 25$: $\mathrm{step}(1) = 1$, $\mathrm{step}(25) = 19$). We record this because a finite-residue reduction is a recurring error here: any argument reducing the dynamics to finitely many residue classes is incorrect, and checking only the representatives $x = 1, 3, \dots, M - 1$ proves nothing — $x = 1$ passes where $x = 25$, in the same class, fails.

6 A machine-certified descent atlas, [P-Lean]

Call $n \equiv r \pmod{2^j}$ a *descent class* if there is $t = t(r, j)$ with $\mathrm{step}^t(n) < n$ for every n in the class. Since the first j binary digits determine the first valuations, such a statement unrolls into a finite algebraic identity and is provable mechanically. Classical examples: $n \equiv 1 \pmod 4$ ($t = 1$), $n \equiv 3 \pmod{16}$ ($t = 2$), $n \equiv 11, 23 \pmod{32}$ ($t = 3$).

Pipeline. (i) an exact tree walk enumerates the *minimal* descent classes level by level (no sampling, no floating point); (ii) each is compiled to a self-contained Lean 4 proof by orbit unrolling; (iii) a certifier runs `lake build` and inspects `#print axioms`, accepting a class *only if* the kernel does, with no `sorry` and only the ambient axioms. Current certificate:

$$\texttt{n_classes} = 197, \qquad \texttt{n_theorems} = 1399,$$
$$\texttt{lake_build_ok} = \texttt{true}, \quad \texttt{axioms_kernel_only} = \texttt{true}.$$

The pipeline re-derives the classical classes, which serves as a regression test.

Remark 6.1 (The certifier is the point)**.** Steps (i)–(ii) can and do emit plausible-looking candidates that are wrong; only (iii) decides. An unchecked generator will happily produce a document asserting a theorem that does not compile. Making the kernel — not the generator — the sole authority is the discipline that prevents this.

Exact descent densities, [V]. The fraction $d(K)$ of odd residues descending within K accelerated steps is an exact dyadic rational:

$$d(1) = \tfrac{1}{2}, \quad d(2) = d(3) = \tfrac{3}{4}, \quad d(4) = d(5) = \tfrac{7}{8}, \quad d(6) = d(7) = \tfrac{55}{64},$$
$$d(8) = \tfrac{115}{128}, \quad d(9) = \tfrac{235}{256}, \quad d(10) = \tfrac{955}{1024}.$$

These are exact (no sampling); the underlying density statement is the classical stopping-time result of Terras [4], so we report the values as a re-computation and claim no new theorem.

7 The intrinsic ceiling

We must first fix the notion precisely, since "resistant" admits several inequivalent readings and the counts depend on which one is used.

Definition 7.1. Run the symbolic orbit of the class $r \bmod 2^j$ using *only* the j known bits: at each step the valuation $\mathrm{v}_2(3x+1)$ is read off, and the available precision drops by that amount. The class is *resistant at level j* if the precision is exhausted before the orbit drops below r — i.e. the j bits do not decide descent. Let $\rho(j)$ be the fraction of odd residues mod 2^j that are resistant.

j	resistant classes	$\rho(j)$ exact	$\rho(j)$
10	91	91/512	0.1777
15	1,610	805/8192	0.0983
20	31,732	7933/131072	0.0605

Table 1: Resistant classes under Definition 7.1, computed exactly (no sampling). The resistant set thins in density but grows without bound in absolute size.

The full exact sequence of resistant counts for $j = 3, \ldots, 20$ is

4, 6, 9, 13, 21, 34, 53, 91, 155, 283, 509, 876, 1610, 2905, 5019, 9247, 16742, 31732.

The successive ratios $\rho(j{+}1)/\rho(j)$ do *not* settle: they oscillate $(0.75, 0.75, 0.72, 0.81, 0.81, 0.78, 0.86, \ldots)$ with an apparent period of about 3, which is what one expects from the irrational rotation by $\log_2 3 \approx 1.585$. Consistent with this, the sequence satisfies *no* linear recurrence with constant coefficients of order ≤ 5 and is not exactly geometric — both checked over the exact rationals. We therefore report the values and refrain from fitting a geometric law to them.

Proposition 7.2 (Ceiling, **[P]**)**.** *Enumerating descent classes establishes the conjecture on a set of odd n of density $1 - \rho(j) \to 1$, hence for* almost all *n — the Terras regime [4], sharpened by Tao [5] — but for no finite j does it cover all n. No finite enumeration of residue classes can prove the Collatz conjecture.*

Together with Proposition 5.1 this delimits the method sharply: not only does enumeration fail to terminate, no fixed 2-adic precision even suffices to define the step.

8 Conjecture

Conjecture 8.1 (Champion, **[C]**)**.** Let R be the normalised observable defined along the orbit of n by $R_0 = 0$, $R_{k+1} = (3R_k + 1)/2^{v_k}$. Then $\sup_k R_k$ is maximised, over all odd $n \geq 3$, at $n = 993$ and its doublings $993 \cdot 2^k$.

Consistent with an exhaustive scan; it is a universal statement over an infinite set and is *not* proved. Reported as a conjecture only.

What is not claimed

- No proof of the Collatz conjecture.
- No finite-automaton or finite-residue reduction (Proposition 5.1 shows this is impossible).
- No claim that the champion conjecture is a theorem; it is checked on a finite range only.

Reproducibility

All counts are regenerated by `tools/verify_discoveries.py` and `tools/verify_for_papers.py`; the Lean development builds with `lake build`, and the certifier rewrites a machine-readable certificate.

Acknowledgement

AI-assisted; every claim was verified by exact computation or by Lean's kernel before inclusion, and every claim that failed verification was removed.

References

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